

Week 17 Homework: Use PCA and Explore a Research Proposal

Goal: build a geometric intuition for PCA, use it correctly with a library, and spend most of the week discovering a motivating and feasible research project.

Estimated effort. About 45–60 minutes for PCA and 2–3 hours for guided research-proposal exploration. The proposal is an early draft. It is expected to change after conversations with the mentor.

Deliverables. Submit (1) one completed Python notebook or file named `week17_pca.py`, (2) one labeled 2D PCA plot, (3) the short PCA reflection, (4) a three-idea exploration sheet, (5) a stakeholder and resource map, (6) research proposal v0.1, (7) a Week 17–22 roadmap, and (8) the Feynman Technique post.

AI policy. You may use AI to explain documentation, suggest search terms, help interpret an error message, or ask questions about an unfamiliar research area. You must personally choose the project, verify factual claims and sources, explain the PCA plot, and write the proposal in your own words. If AI is used, add a short note describing what it helped with and what you checked yourself.

Part I: PCA—Understand the Picture and Use the Tool

This assignment does **not** require deriving eigenvalues, implementing PCA from scratch, or calculating a covariance matrix by hand. The goal is to understand the geometry and use a trusted implementation correctly.

Step 1: Watch for geometric intuition

Watch the mentor-provided **Eigenfaces** video, which shows how many image pixels can be represented using a smaller collection of important directions. If that video is unavailable, use either of these introductions:

- [3Blue1Brown: Eigenvectors and Eigenvalues](#) for geometric intuition;
- [StatQuest: PCA Main Ideas in Only Five Minutes](#) for the direct PCA connection; or
- [StatQuest: PCA Step-by-Step](#) if you want a longer explanation.

While watching, sketch one picture showing a cloud of points and a direction through the cloud. Then answer in one or two sentences each:

1. What does it mean to project a point onto a line?
2. Why does PCA choose the direction along which the data are most spread out?
3. If 20 original features become two principal components, what was compressed?
4. Why can a 2D PCA plot be useful even though it loses some information?
5. In the Eigenfaces example, what does one principal component visually represent?

Step 2: Follow the guided PCA example

Use the built-in Wine dataset so that data collection does not distract from learning PCA. Type and run the following code. Replace each `TODO` with a short comment explaining what that line does.

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load 13 numerical measurements for each wine sample.
data = load_wine(as_frame=True)
X = data.data
y = data.target

# TODO: explain why scaling is useful before PCA here.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# TODO: explain why n_components=2 is useful for visualization.
pca_2d = PCA(n_components=2)
X_2d = pca_2d.fit_transform(X_scaled)

print("Original shape:", X.shape)
print("PCA shape:", X_2d.shape)
print("Explained variance:", pca_2d.explained_variance_ratio_)
```

```

print("Total shown in 2D:", pca_2d.explained_variance_ratio_.sum())

plt.figure(figsize=(7, 5))
for class_id, class_name in enumerate(data.target_names):
    mask = (y == class_id)
    plt.scatter(X_2d[mask, 0], X_2d[mask, 1], label=class_name,
                alpha=0.75)

plt.xlabel(
    f"PC1 ({pca_2d.explained_variance_ratio_[0]:.1%} variance)"
)
plt.ylabel(
    f"PC2 ({pca_2d.explained_variance_ratio_[1]:.1%} variance)"
)
plt.title("Wine Data Projected onto Two Principal Components")
plt.legend()
plt.tight_layout()
plt.show()

```

Make sure the code runs and save the plot. Then answer:

1. How many original features are there? How many numbers describe each sample after the 2D transformation?
2. What percentage of the total variance is visible in the two displayed components?
3. What does one dot represent? What do the colors represent?
4. Describe one visible pattern without claiming that PCA proved why the pattern exists.
5. Name one kind of information that may be hidden outside this 2D view.

Step 3: Let the library choose a useful compressed size

Add this short experiment:

```

# Keep enough components to explain at least 90% of the variance.
pca_90 = PCA(n_components=0.90)
X_90 = pca_90.fit_transform(X_scaled)

print("Components needed for 90% variance:", pca_90.n_components_)
print("Compressed shape:", X_90.shape)

```

In three to five sentences, explain the tradeoff between using two components for an easy picture and using more components to preserve more of the original variation. No manual eigenvalue calculations are required.

Part II: Explore the Research Proposal

The revised program ends at Week 22. Starting now, the student and mentor will develop one small but complete demo research project. This week's purpose is exploration, not pretending that the first idea is already perfect.

Step 1: Notice what creates excitement

Complete at least eight of these sentence starters quickly. Do not judge the ideas yet.

- I keep wondering why ...
- I would love to build a small tool that ...
- Something inconvenient or unfair that I notice is ...
- A smart-grid or AI idea I want to see with my own eyes is ...
- A claim I have heard but do not completely believe is ...
- A dataset, device, or program I would enjoy exploring is ...
- A problem affecting people I care about is ...
- A result that would surprise me is ...
- A technique from an earlier week that I want to use again is ...
- If failure were acceptable, I would try ...

Circle the three responses that create the strongest curiosity or excitement. Turn each one into a short project seed using this frame:

I am interested in _____ because _____. I might investigate this
by _____.

Step 2: Explore three seeds without forcing a decision

For each seed, complete the table. Short phrases are enough.

Prompt	Seed 1	Seed 2	Seed 3
What excites me?			
Who might care?			
What could I observe or measure?			
What data or materials might exist?			
Who could advise me?			
Biggest obstacle			
Small result possible in six weeks			

Do one small reality check for every seed. This may be a 10-minute literature search, opening a candidate dataset, checking whether a library exists, or asking a knowledgeable person one question. Record what you learned and the source or person consulted.

Step 3: Have a mentor conversation

Bring the three seeds to the instructor or mentor. Do not begin by asking, “Which one is best?” First explain the excitement behind each seed. Record the mentor’s response to these questions:

1. Which seed has the clearest observable problem?
2. Which seed has resources that can be accessed immediately?
3. Which seed is too large, and how could it be reduced?
4. What important assumption should be checked next?
5. Who else might be a useful stakeholder or adviser?

After the conversation, choose one **working seed**. This is a reversible choice. Write one paragraph explaining why it is worth exploring next and what evidence could make you change direction.

Step 4: Move from a topic to a researchable question

Use the following guided sequence. It is acceptable to write “not known yet” and turn that uncertainty into a next action.

1. **Topic:** What general area interests you?
2. **Observation:** What specific behavior, difficulty, pattern, or uncertainty did you notice?
3. **Stakeholder:** Who experiences the problem or might use the result?
4. **Why it matters:** What decision, cost, risk, or understanding could improve?
5. **Current evidence:** What source suggests that the problem is real?
6. **Research question:** What smaller question could you answer with available evidence?
7. **Baseline:** What simple existing method, average, rule, or comparison will prevent an impressive-looking but meaningless result?
8. **Smallest complete result:** What is the minimum result that would allow an honest conclusion?
9. **Boundary:** What people, locations, data, methods, or claims are outside this project?

Use this optional sentence frame:

In *[context]*, *[stakeholder]* faces *[specific problem]*. I will investigate whether *[method or factor]* changes or explains *[measurable outcome]*, using *[data or evidence]* and comparing it with *[baseline]*.

Step 5: Map stakeholders, resources, and requests

Identify at least three stakeholders. One should be affected by the problem, one should have relevant expertise, and one should control a resource or permission. For each, answer:

- What might this person care about?
- What could this project ask from them?
- What evidence or explanation would help them respond?
- How could the project benefit or inconvenience them?

Write one realistic request for the coming week. Examples include asking for a 20-minute advice meeting, requesting a dataset description, checking a measurement definition, or receiving feedback on the scope. Avoid vague requests such as “Please support my project.”

Research Proposal v0.1

Write one to two pages. This is a conversation document, not a funding application. Use plain language and include:

1. **Working title and spark:** What makes you want to do this?
2. **Problem and stakeholder:** What is happening, to whom, and why might it matter?
3. **Working research question:** What exactly might the project learn or demonstrate?
4. **Evidence available now:** What data, sources, tools, skills, and advisers can you access?
5. **First method and baseline:** What is the simplest credible comparison?
6. **Smallest complete result:** What can definitely be finished and shown?
7. **Unknowns and risks:** What still needs to be checked?
8. **Stakeholders and next request:** Who should be consulted, and what will you ask?

Mark statements using these labels where helpful:

- **Known:** supported by a checked source or direct observation;
- **Assumption:** plausible but not yet checked; and
- **Question:** requires literature, data, an experiment, or expert advice.

Flexible Week 17–22 Roadmap

Fill in one achievable output for each week. Weeks 18 and 19 should reduce uncertainty before the project becomes complicated.

Week	Purpose	My next visible output
17	Explore and choose a working proposal seed	
18	Check literature, data access, and definitions	
19	Make the smallest end-to-end demonstration	
20	Run the main comparison or experiment	
21	Review errors, limitations, and feedback	
22	Present the result, artifact, and reflection	

Add two fallback decisions:

- If the preferred data or resource is unavailable by the end of Week 18, I will ...
- If the first method does not work by the end of Week 19, I will simplify the project by ...

Feynman Technique Post

Create one public-facing learning note titled “**PCA: Turning Many Measurements into a Picture.**” Include:

- an original sketch of a point cloud and its first principal direction;
- a plain-language explanation of projection and why PCA can compress data;
- the 2D PCA plot from the guided example;

- one warning that a 2D projection does not show everything; and
- one sentence about how an exciting observation can become a research question.

Submission checklist. The PCA code runs with scikit-learn; the plot has labeled axes and explained-variance percentages; the student can explain projection without deriving eigenvalues; three research seeds were explored; at least one reality check and one mentor conversation were recorded; the proposal separates known facts, assumptions, and questions; the six-week plan begins with a small demonstration; and any AI assistance is disclosed.